

PMUTs Meet Wearables: Miniaturized, Battery-Friendly and performant ultrasound

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The three most important characteristics of ultrasonic transducers for wearables are miniaturization, high sound pressure, and high sensitivity—all at low, battery-compatible voltages. This paper presents piezoelectric micromachined ultrasonic transducers (PMUTs) designed to address the challenges of wearable devices. PMUTs, manufactured using microfabrication technologies, can be highly miniaturized and integrated into wearable applications, such as patches. PMUTs can deliver both the transmit power required for cavitation and the high signal-to-noise-ratio (SNR) needed, for example, in optoacoustics.